Notes on Collapses and Persistent Path Homology

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Abstract

These notes record what does, and what does not, transfer from the collapse-based simplification of simplicial persistent homology to the path homology of digraphs.

1 Preliminary Concepts

Definition 1.1. A *digraph* is a pair $G = (V, E)$ where $V = \{v_1, \ldots, v_n\}$ is a finite set of *vertices* and $E \subseteq V \times V$ is a set of *arcs* (ordered pairs of distinct vertices).

2 Elementary Collapses

Let $\sigma^{(n-1)} < \tau^{(n)} \subset K$ and $L = K \backslash \{\sigma, \tau\}$ be the path complex resulting from deleting σ and τ from K. We say that K collapses onto L by an elementary collapse of dimension n. We denote by $K \searrow L$ if K can be transformed into L by a finite sequence of elementary collapses.

The collapse $K^\omega \searrow L^\omega$ preserves weighted homology if $H_i(K^\omega) \cong H_i(L^\omega)$, $\forall i \geq 0$. In the article (Wu et al., 2020) was shown that weighted simplicial collapses do not necessarily preserve weighted homology. Nonetheless, (Wang et al., 2019) showed that persistent homology with field coefficients is independent on weights but it will depend on the weights if the coefficients are in a general ring R (e.g., $\mathbb{Z}$).

3 Strong collapses and persistent homology

Definition. A vertex v in K is called a dominated vertex if the link of v in K, $lk_K(v)$, is a simplicial cone, that is, there exists a vertex $v' \neq v$ and a subcomplex

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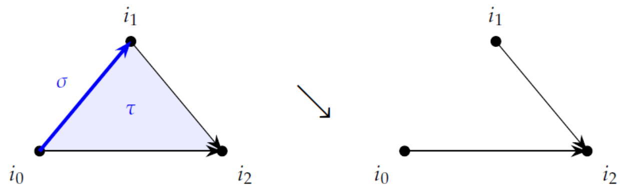

Figure 1: Elementary collapses. In the path complex of the triangle $i_0 \to i_1 \to i_2$, $i_0 \to i_2$, the arc $\sigma = i_0 i_1$ has the single coface $\tau = i_0 i_1 i_2$, which is maximal; removing the pair is an elementary collapse, and here it does preserve path homology.

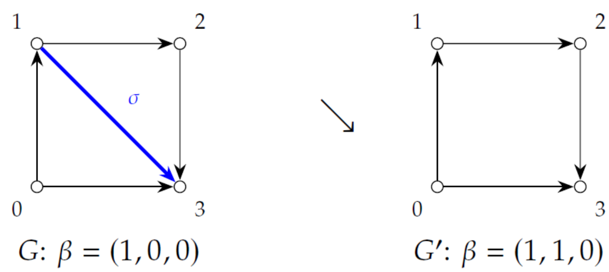

Figure 2: Example **??**. Collapsing the free pair (13, 013) turns a contractible digraph into the non-contractible 4-cycle.

L in K, such that $lk_K(v) = v'L$. An elementary strong collapse is the deletion of a dominated vertex v from K, which we denote with $K \searrow\searrow^e K \backslash v$.

In (Pritam, 2020) was showed that the persistence homology of a filtration of simplicial complexes is preserved under strong collapse, i.e., given a filtration of simplicial complexes K_i connected through inclusion maps

$$Z : \{K_1 \xrightarrow{f_1} K_2 \xrightarrow{f_2} K_3 \xrightarrow{f_3} \dots \xrightarrow{f_{(n-1)}} K_n\}, \tag{1}$$

we independently strong collapse the complexes of the filtration to reach a filtration

$$Z^c : \{K_1^c \xrightarrow{f_1^c} K_2^c \xrightarrow{f_2} K_3^c \xrightarrow{f_3^c} \dots \xrightarrow{f_{(n-1)}^c} K_n^c\}. \tag{2}$$

References

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